
Software Design Specification

for

Ultimate Athletic Gym Management System

Version 1.0

Prepared by

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Course: TME 3413 Software Engineering Lab

Lab Section: *Group 1*

Date: 6 January 2023

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed

1 Introduction

Here Castor is going to define software design specifications for Ultimate Athletic Gym Management System. Centered on the previous Software Requirements Specification, this document emphasizes the designing requirements. In the following pages, we specify the purpose of this document, the overview of the Ultimate Athletic Gym Management System and the references used in the production of this document.

1.1 Purpose

This Software Design Specification (SDS) is used to report our Ultimate Athletic Gym Management System product strategy. This document's goal is to provide a high-level overview of the design analysis as well as a low-level structure for the Ultimate Athletic Gym Management System Software framework. The entirety of the design depends on the functions, capabilities, imperatives and what's more, the interfaces of the framework. Clients can manage the gym's membership and any important input through this web application. This SDS will manage the reader in detail for our product plan of the web application.

1.2 System Overview

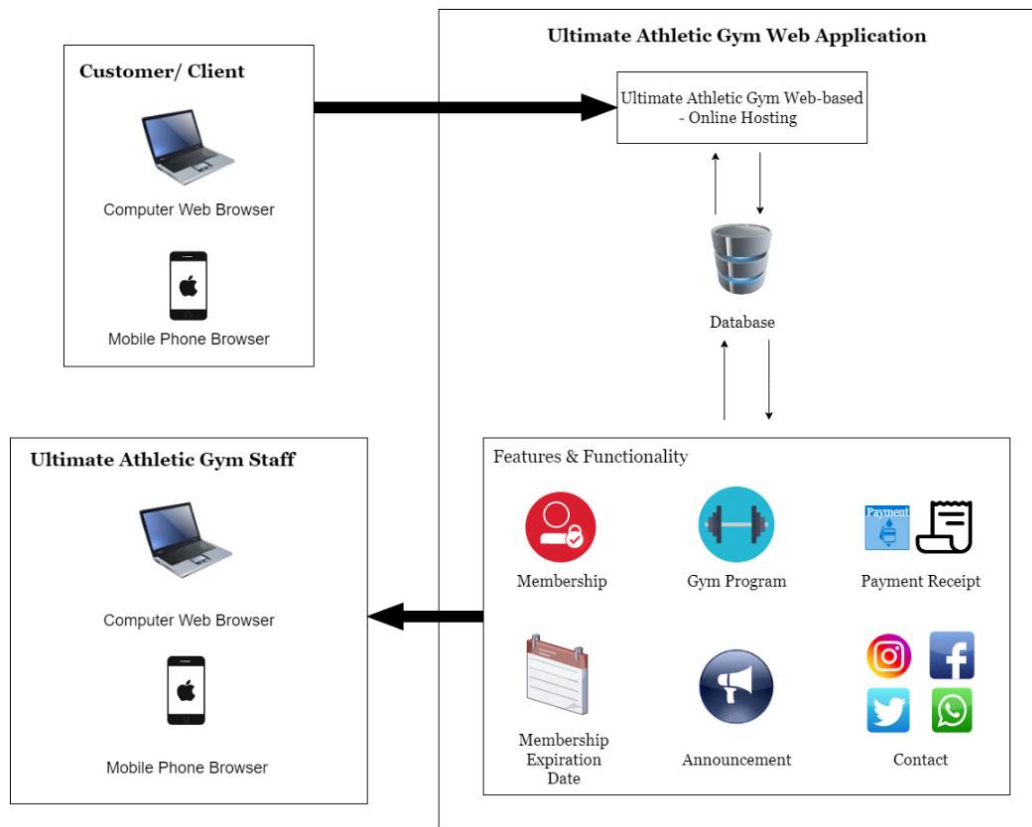


Figure 1: System architecture of Ultimate Athletic Gym Management System

Ultimate Athletic Gym staffs and customers can access the web-based system via devices such as PCs, laptops, tablets, or even smartphones, as long as they have a built-in web browser and a stable internet access, but each user would have a various version of functionality as each user has different requirements in Ultimate Athletic Gym Web Application. Membership registration, customer payment, customer personal tracker, gym special program, gym services and facilities, company background and Ultimate Athletic Gym contact information are all included in the Ultimate Athletic Gym Web Application customer view while for the administrator/staff, it would have an advance functionality compare to the customer as they can perform an addition, reading, updation, and deletion which allows the administrators to manage customers data and record

details. Administrators can add, view, edit, and delete programs details, services and facilities, and membership prices as well as company information, from the database. Administrators may also view the membership expiry date and the history of e-invoices while updating the progress, session, and routine trackers.

1.3 Definitions, Acronyms and Abbreviations

Acronyms and Abbreviations;

- CSS - Cascading Style Sheet
- HTML - Hypertext Markup Language
- IEEE – Institute of Electrical and Electronic Engineers
- MySQL - Structured Query Language
- PHP – Hypertext Preprocessor
- SRS - Specific Requirement Specification
- UI - User Interface
- XAMPP - Cross-Platform (X), Apache (A), MariaDB (M), PHP (P), and Perl (P)

1.4 Supporting Materials

For this document, the IEEE formatting requirements are used. Formatting requirements involve the document to be in Arial Font to be in size 11, single spaced and 1” margins at all sides. Bold letters indicate the main topic and the subtopic for ease of readability of topics desired. All the diagrams and figures included in this document are numbered accordingly in order and a short description is provided where necessary.

[1] P. Mirenda, "Software Design Specification Document," 17 March 2006. [Online]. Available: <https://lostclouds.com/2Communicate/project/SDS.pdf>. [Accessed 15 December 2022]

1.5 Document Overview

This SDS is a document created expressly for the Ultimate Athletic Gym Management System design process. This SDS is divided into four sections: Introduction, Architecture, High Level Design, and Appendix. This SDS will be briefly described in the Introduction,

as will the first section. The second element, Architecture, defines the overall design of the system and serves as the foundation for further detailed design work. The High-Level Design part expands on the topics covered in the Architecture section. Typically, this section will be divided into separate documents for each design area. Minutes of group meetings, group activities, and other pertinent information are often included in the Appendix section, allowing our instructor to assess the efforts made to compile this report.

It is recommended that our instructor read this Software Design Specification (SDS) from beginning to end. We advised our client to study only Introduction, Architecture, and High-Level Design.

2 Architecture

In this section, the conceptual model and logic behind the Ultimate Athletics Gym Management System is shown. Component diagram is used to describe each aspect of the relationship between each component and the dependencies of the component. The Ultimate Athletics Gym Management System can be classified architecturally in three layers:

1. Interface layer (User interface)
2. Application layer
3. Storage layer (Database)

The interface layer provides a graphical user interface via which users can interact with the application or software. The use of web languages such as HTTP, CSS, and PHP can be applied as the interface layer framework. The logic and guidelines for data management in the storage layer are included in the application layer, which can also be recovered depending on the requirements of the system's user. This application layer allows the admin to manage the data file access and houses the parses for the data files.

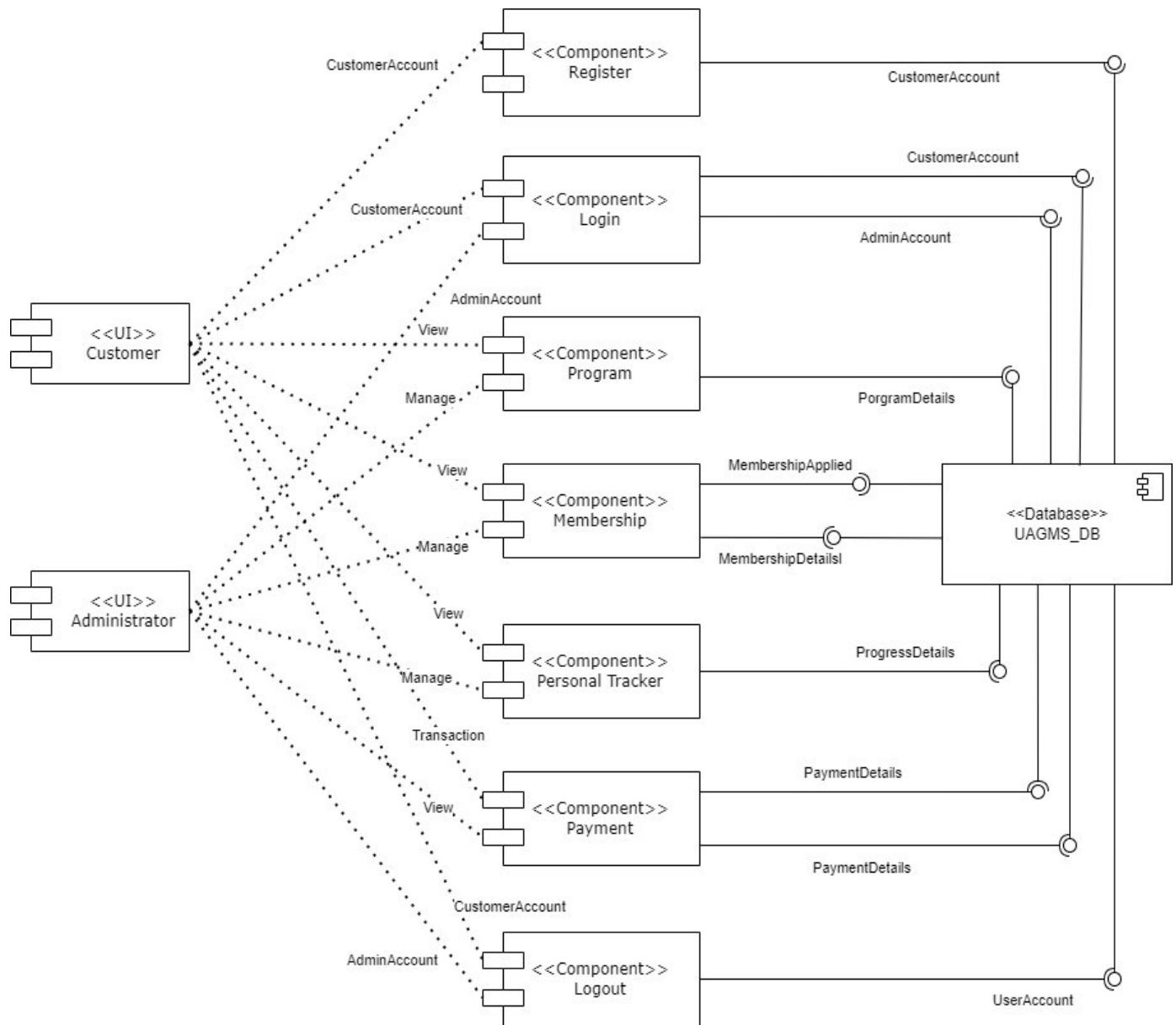


Figure 2: Component Diagram of Ultimate Athletics Gym Management System

The data required by the system will be kept in the storage layer (database). More information on each component is covered in Section 2.2. In the following section, the components and description of the discussed design are described.

2.1 Overview

Based on the figure 2, the components of the system are:

- Customer
- Administrator
- Register
- Login
- Program
- Membership
- Program
- Payment
- Database
- Logout

2.2 System Component

i. 2.2.1 Customer

The customer user interface is a component of a graphical interface that is accessible for the end user. The website provides the customer with access to information regarding the client company background, program, membership and services.

ii. 2.2.2 Administrator

The Administrator user interface is a component that is accessible for the end user (admin). This graphical user interface allows only the admin to manage the program, progression, membership, approval status.

iii. 2.2.3 Register

The registration feature is designed for customers and clients to create their account by filling in the details that the company needs to complete their application. Customers are required to key in their first name, last name, email address, and password to continue with using the website.

iv. 2.2.4 Login

The login feature is created for the client to log into the system to manage the content of the system for the customer to view their membership, services, program, and progress. Admin is required to login to enter the administration interface. Existing customers can view their profile, membership, program, and progression.

v. 2.2.5 Program

The program component is used for customers to view and apply the gym program. The customer is required to log in before applying for any gym program. The admin will access and monitor the list of programs and the list of customers who apply for the program and approve their application. All the program application approval will be stored in the database

vi. 2.2.6 Membership

The membership component is made for the customer to apply and view their membership status. The customer is required to log in before applying or viewing the membership. The administrator is able to oversee and manage the list of membership applications and do approve of the application. All the membership detail will be saved in the database

vii. 2.2.7 Payment

The purpose of the payment feature is for the customer to make transactions. The website gives the customer the option to make payment through Online Banking or cash. The customer is required to fill in the required information for record purposes. The customer will submit the proof of payment on the website for the admin to approve their payment and send the receipt to the customer's email.

viii. 2.2.8 Database

The UAGMS_DB is the name of the database that stores all the data required for the Ultimate Athletics Gym Management System to operate.

ix. 2.2.9 Logout

This function enables the admin and customer to end their session of using the website by logging out of the system.

3 High Level Design

The software product's potential architecture is described in high-level design (HLD). The architectural diagram is a high-level picture of a system, outlining its major components and the connections between them. Customer and Admin state charts are used to demonstrate the dynamic behavior of the Ultimate Athletic Gym Management System. Both the consumer and administrator state diagrams describe behavior. The Ultimate Athletic Gym Management System is depicted in its entirety in Figures 3 and 4. The system is accessible by anyone with access to the Internet and a web browser by virtue of being housed on a publicly accessible server.

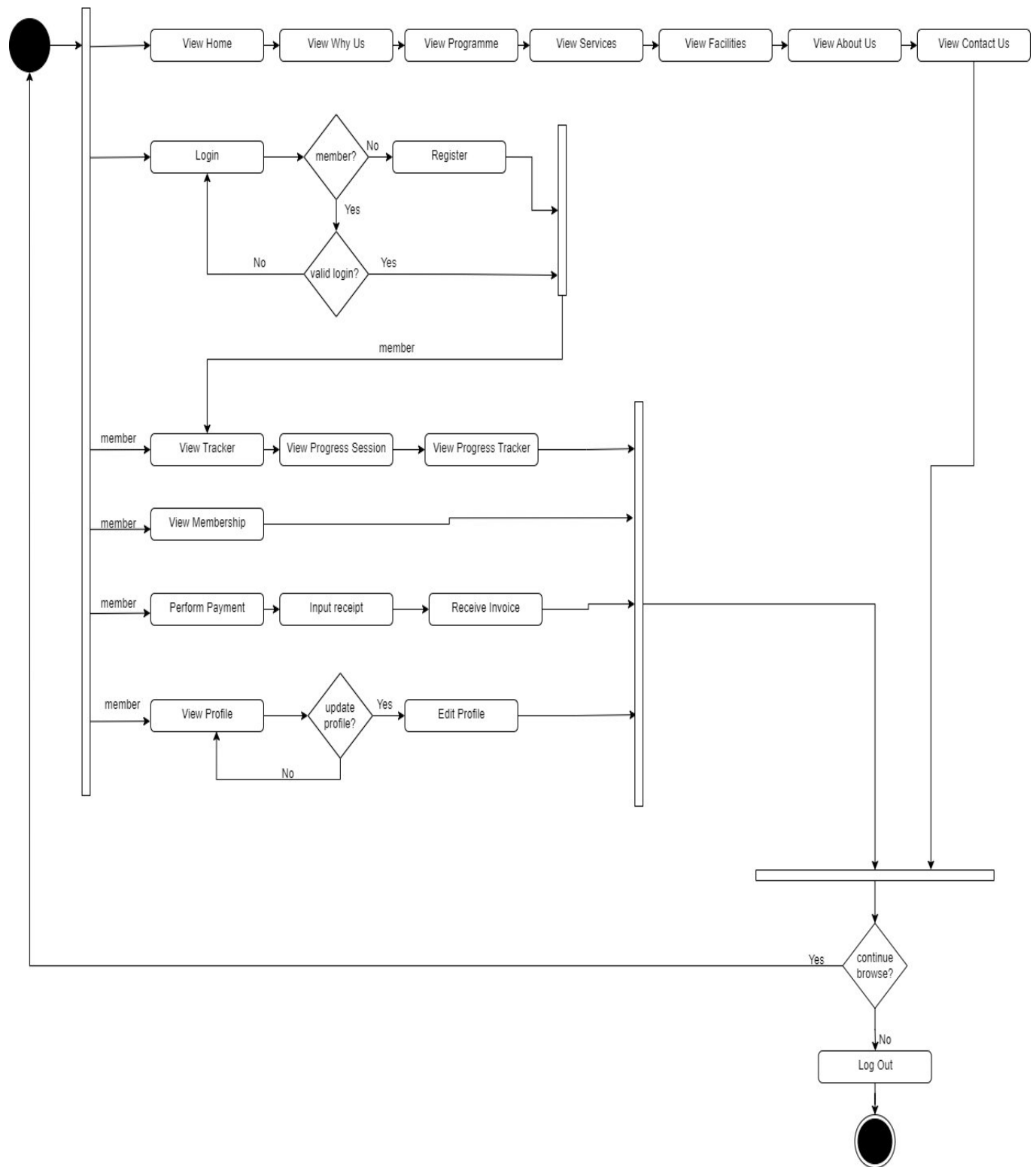


Figure 3: Customer state chart – overall view of Ultimate Athletic Gym Management System using state chart through the customer entity

The Ultimate Athletic Gym Management System website provides a specific set of events to customers who have opened the website.

1. The user can view the Home Page. At this page, users can view Why Us, view Programmer, view Services, view Facilities, view About Us, and view Contact Us by click button at nav bar or can scroll down the page.
2. The user can login to the website. If the user is non-member, they can register for the first-time user. The users need to re-login if the login is not valid.
3. The member users can view the tracker. At this page, users can view progression sessions and progress tracker.
4. The member users can view the membership details by clicking the “Membership” button.
5. The member users can perform payment. They need to insert a screenshot receipt after payment. After that, the user will receive an invoice.
6. If the member users want to view or edit their profile by clicking the “Profile” button.

When a user makes a payment, they have to fill out the payment details in order for the payment to go through. The payment is made outside of the system, where the user can pay with cash, a bank transfer, or a QR code. When the user has already filled out the details, the system will make an invoice and let the user print it at the same time.

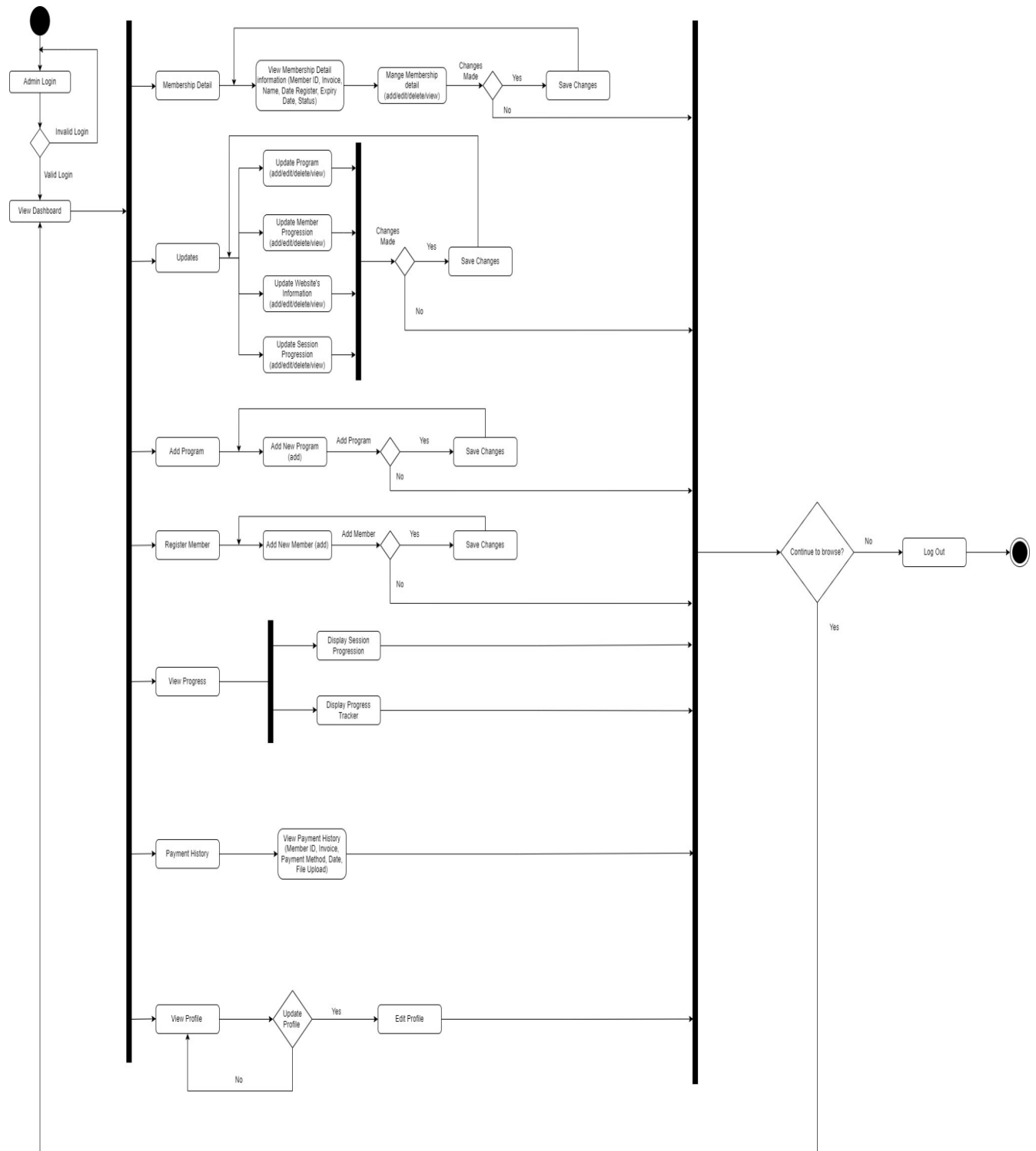


Figure 4: Admin state chart - overall view of Ultimate Athletic Gym Management System using state chart through the administrator link.

In the event of the administrator successfully landing on the admin page of Caker Holic Web Application, the administrator must be logged in first for verification before proceeding to the admin dashboard. After successfully logged in, the admin will be redirected to the dashboard page. There are few events can be triggered:

1. Admin can view Membership details and update membership details.
2. Admin can do update functions in update pages, such as update program, update member progression, update website's information, and update session progression.
3. Admin can perform an add function which is to add a new program on the website.
4. Admin can register new members on the register member page.
5. Admin can view session progression and tracker session on view progress page by entering the members id.
6. Admin can view the payment history that has been made by the customer.
7. If the admin wants to view or edit their profile by clicking the "Profile" button.

On the events that utilize CRUD operations, any changes made will be saved into the database and updated to the dashboard page. The system will successfully be terminated if the administrator logged out from the admin page.

3.1 View / Model Component 1..n

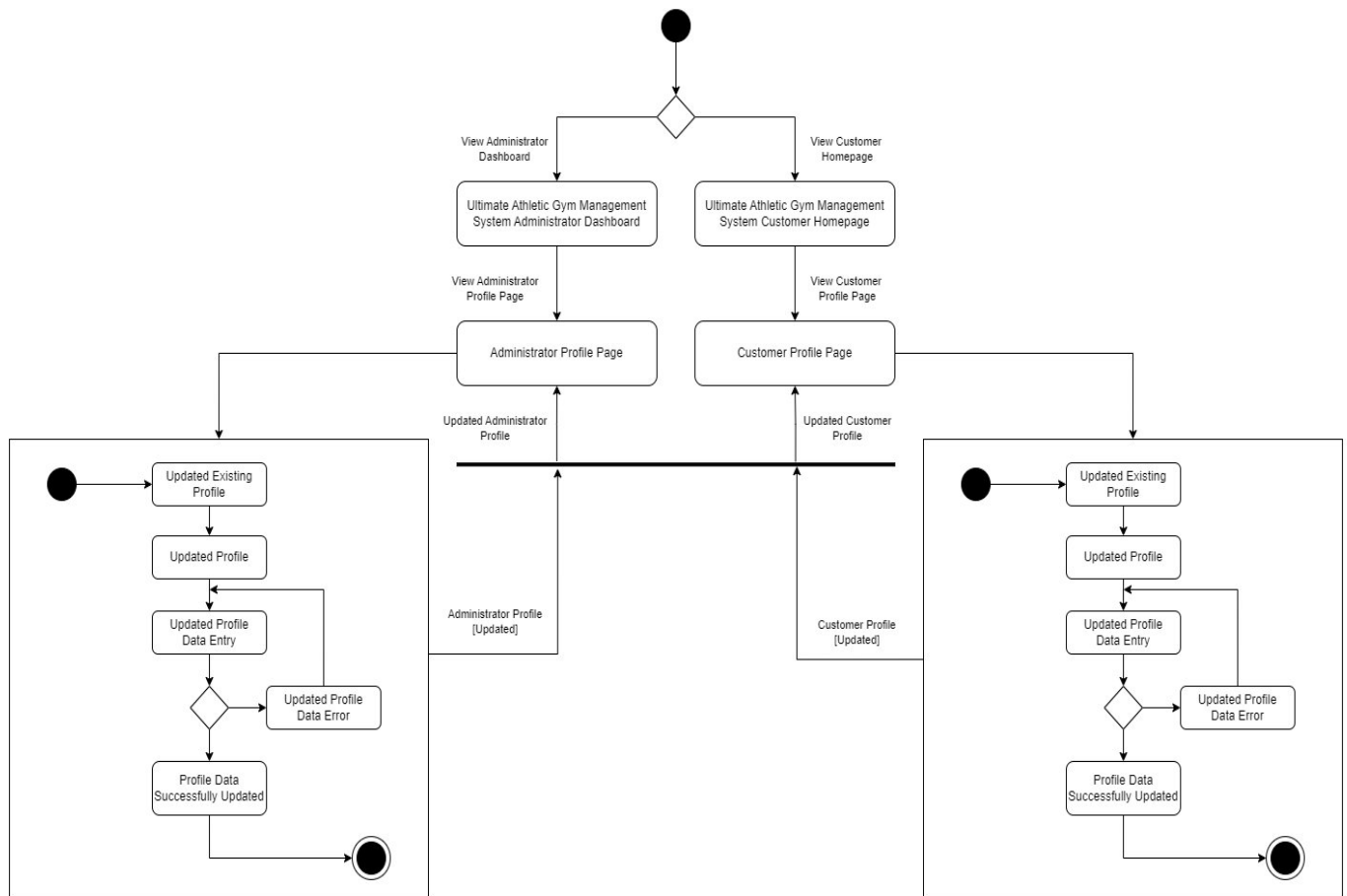


Figure 5: Update Profile state chart model components

If a user has login to the Ultimate Athletics Gym Management System homepage (customers side) successfully, few events are possible to accessed:

1. Customers can edit and update their existing profile information.

When administrators has login to the Ultimate Athletic Gym Management System dashboard (administrators page) successfully, few events can be accessed:

1. Administrators can edit and update from existing profile information.

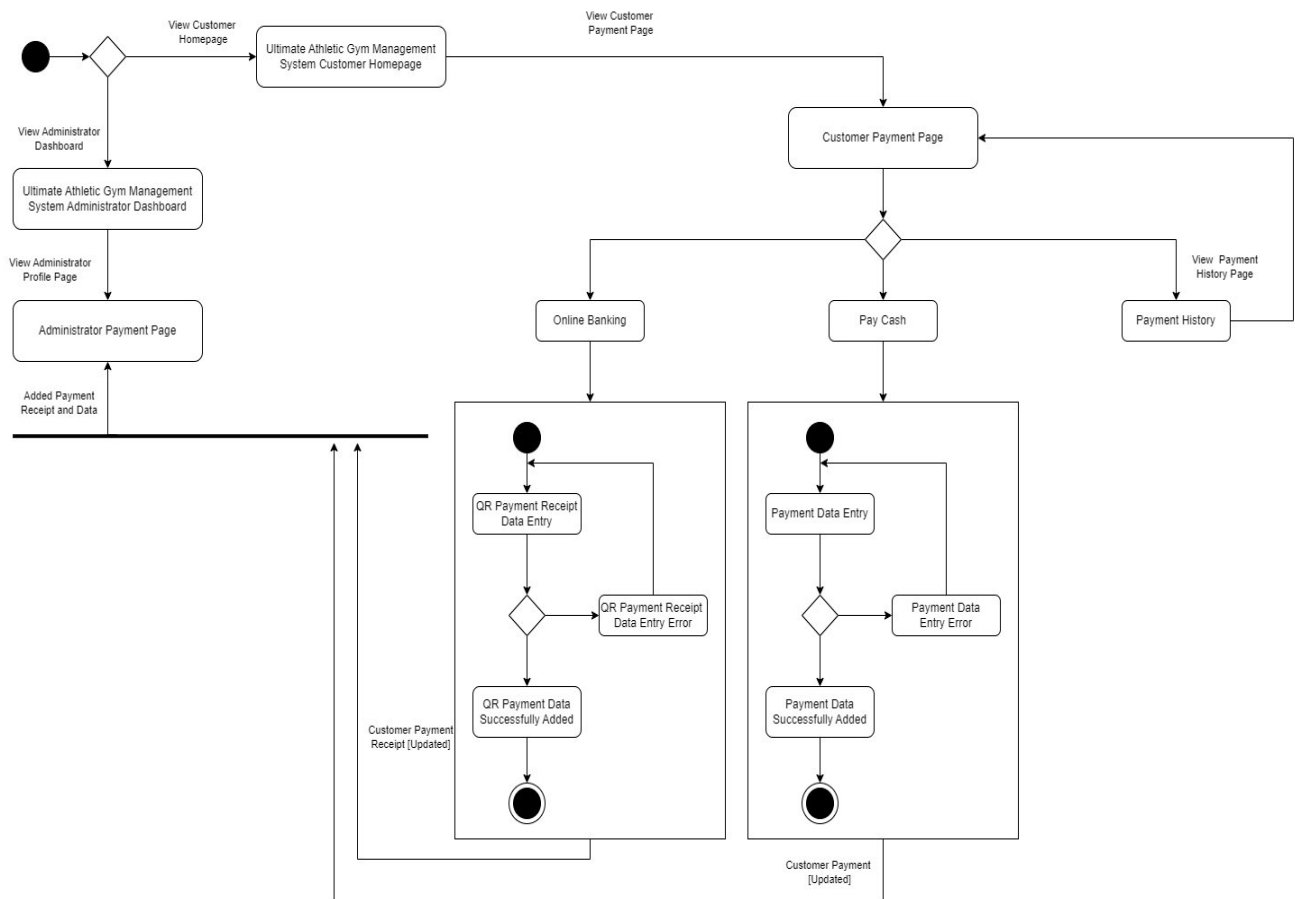


Figure 6: Payment state chart model components

If a user has login to the Ultimate Athletics Gym Management System homepage (customers side) successfully, they can make payment through the Customer payment page. Few events are possible to accessed:

1. Customers can add a new payment receipt with their chosen method.
2. Customers can view payment history

When administrators has login to the Ultimate Athletic Gym Management System dashboard (administrators page) successfully, few events can be accessed:

1. Administrators can view the payment history.

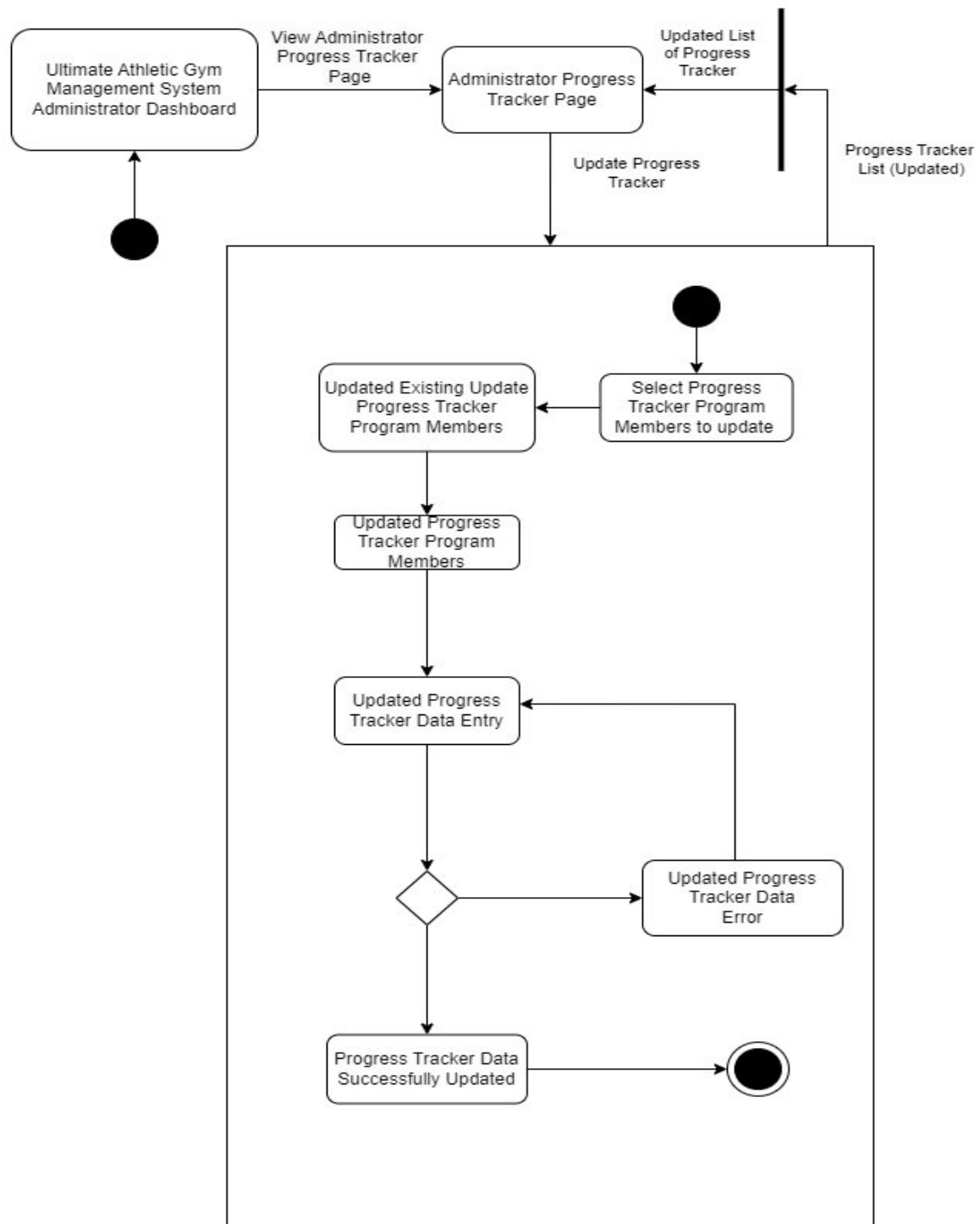


Figure 8: Update Member Progression state chart model components

When administrators has login to the Ultimate Athletic Gym Management System dashboard (administrators page) successfully, administrators can go to update member progression, few events can be accessed:

2. Administrators can view the administrator update member progression page.
3. Administrators can add the data of member progression.
4. Administrators can edit and update the existing data of member progression.
5. Administrators can delete the existing data of member progression.

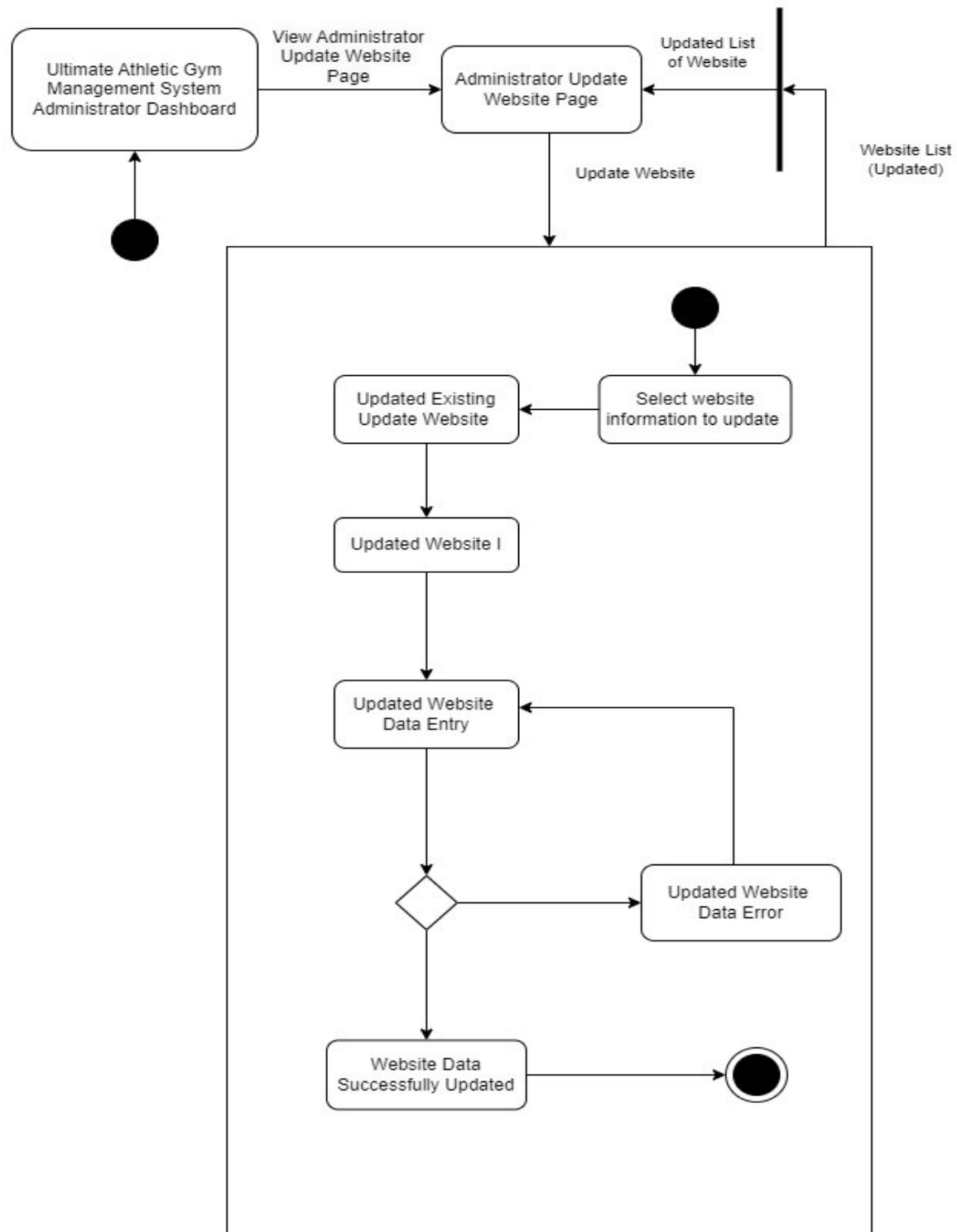


Figure 9: Update Website Information state chart model components

When administrators has login to the Ultimate Athletic Gym Management System dashboard (administrators page) successfully, administrators can go to update website information, few events can be accessed:

1. Administrators can view the administrator update website information page.
2. Administrators can add the website information.
3. Administrators can edit and update the existing data of website information.
4. Administrators can delete the existing data of website information.

6. Appendix A – Group Log

Castor Team Meeting

Meeting Minutes

4th Meeting

Location: WhatsApp Group

Date: 12th December 2022

Time: 9:00pm

Attendees:

Mohammad Amir Alam Bin Rahim Omar (Project & Design Manager)

Azreen Shafieqah Binti Asri (Customer Interface & Design Manager)

Harith Zakwan Bin Zakaria (Customer Interface & Implementation Manager)

Faizatul Fitri Bin Boestamam (Test & Support Manager)

Mohammad Hamka Izzuddin Bin Mohamad Yahya (Planning & Support Manager)

Iman Tarmizi Rosalina (Quality & Process Manager)

Absence with apologies: N/A

At 9:00 PM, Mohammad Amir Alam Bin Rahim Omar welcomed the team members and then called the meeting to order.

Confirmation of the Minutes: Mohammad Amir Alam started the meeting by distributing roles for each team member.	
Topic Discussed	1) System Design Specification report content 2) System Design Specification report task distribution
1.0 Agenda Item	
Discussion	1.1 Mohammad Amir Alam started the discussion by distributing team member roles. 1.1.1 Mohamad Amir Alam was distributing tasks for the System Requirement Specification report. He decided the distribution were introduced by Mohamad Amir Alam, architecture by Iman Tarmizi Rosalina and Faizatul Fitri, high level design by Mohamad Amir Alam, Azreen Shafieqah, Harith Zakwan and Mohammad Hamka.

	1.2 Mohammad Amir Alam was asking everyone if the distribution task was okay or not.
	1.3 Mohammad Amir Alam was giving a summary about the System Design Specification among the team members.
Action Items	-
Adjournment	With no further business to discuss, the meeting was adjourned at 9:30 pm.
Next Meeting Date & Location	To be announced

Table 1: 4rd Meeting Minutes

5th Meeting

Location: WhatsApp Group

Date: 1st January 2023

Time: 1:00pm

Attendees:

Mohammad Amir Alam Bin Rahim Omar (Project & Design Manager)

Azreen Shafieqah Binti Asri (Customer Interface & Design Manager)

Harith Zakwan Bin Zakaria (Customer Interface & Implementation Manager)

Faizatul Fitri Bin Boestamam (Test & Support Manager)

Mohammad Hamka Izzuddin Bin Mohamad Yahya (Planning & Support Manager)

Iman Tarmizi Rosalina (Quality & Process Manager)

Absence with apologies: N/A

Mohammad Amir Alam Bin Rahim Omar opened the meeting by extending a warm welcome to the team members at 1:00 PM.

Confirmation of the Minutes:

Mohd Amir Alam Bin Rahim Omar started the meeting about the submission documentation report's content.

Topic Discussed	1) All documentation report content 2) Progress system
1.0 Agenda Item	
Discussion	1.1 Mohammad Amir Alam started the discussion by distributing team member roles. 1.1.1 Mohamad Amir Alam was distributing tasks for the documentation report. He decided the distribution task.
	1.2 Mohammad Amir Alam was asking everyone if the distribution task was okay or not.
	1.3 Azreen Shafieqah ask all the member about their progress system.
	1.4 Mohammad Amir Alam was giving a summary about the checklist documentation to be submitted among the team members.

Action Items	-
Adjournment	With no further business to discuss, the meeting was adjourned at 2:00 pm.
Next Meeting Date & Location	To be announced

Table 2: 5th Meeting Minutes